```
!pip install spacy_transformers
!pip install -U spacy
```

```
Collecting spacy_transformers
  Downloading spacy_transformers-1.3.9-cp312-cp312-manylinux_2_17_x86_64.manylinu
Requirement already satisfied: spacy<4.1.0,>=3.5.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-pa
Collecting transformers<4.50.0,>=3.4.0 (from spacy_transformers)
  Downloading transformers-4.49.0-py3-none-any.whl.metadata (44 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 44.0/44.0 kB 1.7 MB/s eta 0:00:00
Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: srsly<3.0.0,>=2.4.0 in /usr/local/lib/python3.12/d
Collecting spacy-alignments<1.0.0,>=0.7.2 (from spacy_transformers)
  Downloading spacy_alignments-0.9.2-cp312-cp312-manylinux_2_17_x86_64.manylinux2
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/pyth
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/pyth
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/d
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/d
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.
Requirement already satisfied: weasel<0.5.0,>=0.1.0 in /usr/local/lib/python3.12/
Requirement already satisfied: typer<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in /usr/local/lib/python3.
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: networkx in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in /usr/local/lib/
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in /usr/local/li
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in /usr/local/lib/
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/pyt
Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in /usr/local/lib/pyt
Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in /usr/local/lib/pyth
Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in /usr/local/lib/py
Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in /usr/local/lib/p
Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in /usr/local/lib/p
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/py
Requirement already satisfied: nvidia-nccl-cu12==2.27.3 in /usr/local/lib/python3
Requirement already satisfied: nvidia-nvtx-cu12==12.6.77 in /usr/local/lib/python
Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in /usr/local/lib/p
Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in /usr/local/lib/pyt
Requirement already satisfied: triton==3.4.0 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: huggingface-hub<1.0,>=0.26.0 in /usr/local/lib/pyt
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dis
```

```
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dis
Collecting tokenizers<0.22,>=0.21 (from transformers<4.50.0,>=3.4.0->spacy_transf
  Downloading tokenizers-0.21.4-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/di
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/
Requirement already satisfied: language-data>=1.2 in /usr/local/lib/python3.12/di
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.1
Requirement already satisfied: pydantic-core==2.33.2 in /usr/local/lib/python3.12/
```

```
import spacy
from spacy.tokens import DocBin
from tqdm import tqdm
import json
```

```
spacy.__version__
```

```
'3.8.7'
```

```
!nvidia-smi
```

```
Sat Oct 18 11:22:23 2025
+-----------------------------------------------------------------------------------
| NVIDIA-SMI 550.54.15              Driver Version: 550.54.15      CUDA Version: 1
|-----------------------------------------+------------------------+----------------
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Unc
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Co
|                                         |                        |
|=========================================+========================+================
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |
| N/A   68C    P8             11W /   70W |       2MiB /  15360MiB |      0%
|                                         |                        |
+-----------------------------------------+------------------------+----------------

+-----------------------------------------------------------------------------------
| Processes:
|  GPU   GI   CI        PID   Type   Process name                              GPU
|        ID   ID                                                               Usa
|===================================================================================
|  No running processes found
+-----------------------------------------------------------------------------------
```

```
!git clone https://github.com/laxmimerit/CV-Parsing-using-Spacy-3.git
```

```
Cloning into 'CV-Parsing-using-Spacy-3'...
remote: Enumerating objects: 82, done.
remote: Counting objects: 100% (76/76), done.
remote: Compressing objects: 100% (58/58), done.
remote: Total 82 (delta 14), reused 74 (delta 14), pack-reused 6 (from 1)
Receiving objects: 100% (82/82), 5.62 MiB | 20.18 MiB/s, done.
Resolving deltas: 100% (14/14), done.
```

```
cv_data = json.load(open('/content/CV-Parsing-using-Spacy-3/data/training/train_
```

```
len(cv_data)
```

```
200
```

```
!python -m spacy init fill-config /content/CV-Parsing-using-Spacy-3/data/trainin
```

```
✓ Auto-filled config with all values
✓ Saved config
config.cfg
You can now add your data and train your pipeline:
python -m spacy train config.cfg --paths.train ./train.spacy --paths.dev ./dev.sp
```

```
cv_data[0]
```

```
['Govardhana K Senior Software Engineer  Bengaluru, Karnataka, Karnataka - Email
me on Indeed: indeed.com/r/Govardhana-K/ b2de315d95905b68  Total IT experience 5
Years 6 Months Cloud Lending Solutions INC 4 Month • Salesforce Developer Oracle
5 Years 2 Month • Core Java Developer Languages Core Java, Go Lang Oracle PL-SQL
programming, Sales Force Developer with APEX.  Designations & Promotions  Willing
to relocate: Anywhere  WORK EXPERIENCE  Senior Software Engineer  Cloud Lending
Solutions -  Bangalore, Karnataka -  January 2018 to Present  Present  Senior
Consultant  Oracle -  Bangalore, Karnataka -  November 2016 to December 2017
Staff Consultant  Oracle -  Bangalore, Karnataka -  January 2014 to October 2016
Associate Consultant  Oracle -  Bangalore, Karnataka -  November 2012 to December
2013  EDUCATION  B.E in Computer Science Engineering  Adithya Institute of
Technology -  Tamil Nadu  September 2008 to June 2012  https://www.indeed.com/r/
Govardhana-K/b2de315d95905b68?isid=rex-download&ikw=download-top&co=IN https://
www.indeed.com/r/Govardhana-K/b2de315d95905b68?isid=rex-download&ikw=download-
top&co=IN  SKILLS  APEX. (Less than 1 year), Data Structures (3 years), FLEXCUBE
(5 years), Oracle (5 years), Algorithms (3 years)  LINKS  https://
www.linkedin.com/in/govardhana-k-61024944/  ADDITIONAL INFORMATION  Technical
Proficiency:  Languages: Core Java, Go Lang, Data Structures & Algorithms, Oracle
PL-SQL programming, Sales Force with APEX. Tools: RADTool, Jdeveloper, NetBeans,
Eclipse, SQL developer, PL/SQL Developer, WinSCP, Putty Web Technologies:
JavaScript, XML, HTML, Webservice  Operating Systems: Linux, Windows Version
control system SVN & Git-Hub Databases: Oracle Middleware: Web logic, OC4J
Product FLEXCUBE: Oracle FLEXCUBE Versions 10.x, 11.x and 12.x  https://
www.linkedin.com/in/govardhana-k-61024944/',
 {'entities': [[1749, 1755, 'Companies worked at'],
   [1696, 1702, 'Companies worked at'],
   [1417, 1423, 'Companies worked at'],
   [1356, 1793, 'Skills'],
   [1209, 1215, 'Companies worked at'],
   [1136, 1247, 'Skills'],
   [928, 932, 'Graduation Year'],
   [858, 889, 'College Name'],
   [821, 856, 'Degree'],
   [787, 791, 'Graduation Year'],
   [744, 750, 'Companies worked at'],
   [722, 742, 'Designation'],
   [658, 664, 'Companies worked at'],
```

```
 [658, 664, 'Companies worked at'],
 [640, 656, 'Designation'],
 [574, 580, 'Companies worked at'],
 [555, 572, 'Designation'],
 [470, 493, 'Companies worked at'],
 [444, 468, 'Designation'],
 [308, 314, 'Companies worked at'],
 [234, 240, 'Companies worked at'],
 [175, 198, 'Companies worked at'],
 [93, 136, 'Email Address'],
 [39, 48, 'Location'],
 [13, 37, 'Designation'],
 [0, 12, 'Name']]}]
```

```
def get_spacy_doc(file, data):
  nlp = spacy.blank('en')
  db = DocBin()

  for text, annot in tqdm(data):
    doc = nlp.make_doc(text)
    annot = annot['entities']

    ents = []
    entity_indices = []

    for start, end, label in annot:
      skip_entity = False
      for idx in range(start, end):
        if idx in entity_indices:
          skip_entity = True
          break

      if skip_entity == True:
        continue

      entity_indices = entity_indices + list(range(start, end))

      try:
        span = doc.char_span(start, end, label=label, alignment_mode='strict')
      except:
        continue

      if span is None:
        err_data = str([start, end]) + "   " + str(text) + "\n"
        file.write(err_data)

      else:
        ents.append(span)

    try:
      doc.ents = ents
      db.add(doc)
```

```
      db.add(doc)
    except:
      pass

  return db
```

```
from sklearn.model_selection import train_test_split
```

```
train,test = train_test_split(cv_data, test_size=0.3)
```

```
len(train), len(test)
```

```
(140, 60)
```

```
file = open('error.txt', 'w')

db = get_spacy_doc(file, train)
db.to_disk('train_data.spacy')

db = get_spacy_doc(file, test)
db.to_disk('test_data.spacy')

file.close()
```

```
100%|██████████| 140/140 [00:01<00:00, 96.93it/s]
100%|██████████| 60/60 [00:00<00:00, 115.68it/s]
```

```
len(db.tokens)
```

```
60
```

```
!python -m spacy train /content/CV-Parsing-using-Spacy-3/data/training/config.cf
```

```
ℹ Saving to output directory: output
ℹ Using GPU: 0

========================== Initializing pipeline ==========================
2025-10-18 12:10:27.878765: E external/local_xla/xla/stream_executor/cuda/cuda_ff
WARNING: All log messages before absl::InitializeLog() is called are written to S
E0000 00:00:1760789427.898863   12710 cuda_dnn.cc:8579] Unable to register cuDNN
E0000 00:00:1760789427.904801   12710 cuda_blas.cc:1407] Unable to register cuBLA
W0000 00:00:1760789427.920129   12710 computation_placer.cc:177] computation plac
W0000 00:00:1760789427.920156   12710 computation_placer.cc:177] computation plac
W0000 00:00:1760789427.920160   12710 computation_placer.cc:177] computation plac
W0000 00:00:1760789427.920166   12710 computation_placer.cc:177] computation plac
2025-10-18 12:10:27.924770: I tensorflow/core/platform/cpu_feature_guard.cc:210]
To enable the following instructions: AVX2 FMA, in other operations, rebuild Tens
Some weights of RobertaModel were not initialized from the model checkpoint at rol
You should probably TRAIN this model on a down-stream task to be able to use it f
```

```
✓ Initialized pipeline

============================= Training pipeline =============================
ℹ Pipeline: ['transformer', 'ner']
ℹ Initial learn rate: 0.0
E    #       LOSS TRANS...  LOSS NER  ENTS_F  ENTS_P  ENTS_R  SCORE
---  ------  -------------  --------  ------  ------  ------  ------
  0       0        5212.88   1239.90    0.08    0.05    0.21    0.00
  3     200      234281.75  81758.35   24.63   42.20   17.39    0.25
  7     400       37081.63  32185.51   44.85   66.95   33.72    0.45
 10     600        5091.73  26081.09   47.49   54.75   41.94    0.47
 14     800        7767.14  26235.70   52.05   76.22   39.52    0.52
 17    1000        2242.38  24048.93   56.51   61.36   52.37    0.57
 21    1200        3998.55  23133.14   53.86   72.08   42.99    0.54
^C
```

```
nlp = spacy.load('/content/output/model-best')
```

```
doc = nlp('my name is talha iqbal. I worked at Twitter. I have 3 years of experi
for ent in doc.ents:
  print(ent.text, "    ->>>>> ", ent.label_)
```

```
talha iqbal     ->>>>>  Name
Twitter     ->>>>>  Companies worked at
3 years     ->>>>>  Years of Experience
```

```
!pip install PyMuPDF
```

```
Collecting PyMuPDF
  Downloading pymupdf-1.26.5-cp39-abi3-manylinux_2_28_x86_64.whl.metadata (3.4 kB
Downloading pymupdf-1.26.5-cp39-abi3-manylinux_2_28_x86_64.whl (24.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 24.1/24.1 MB 94.2 MB/s eta 0:00:00
Installing collected packages: PyMuPDF
Successfully installed PyMuPDF-1.26.5
```

```
import sys, fitz
```

```
fname = '/content/CV-Parsing-using-Spacy-3/data/test/Alice Clark CV.pdf'
doc = fitz.open(fname)
```

```
doc
```

```
Document('/content/CV-Parsing-using-Spacy-3/data/test/Alice Clark CV.pdf')
```

```
text = " "
for page in doc:
  text = text + str(page.get_text())
```

```
text = text.strip()
```

```
text = ' '.join(text.split())
```

```
text
```

```
'Alice Clark AI / Machine Learning Delhi, India Email me on Indeed • 20+ years of
experience in data handling, design, and development • Data Warehouse: Data analy
sis, star/snow flake scema data modelling and design specific to data warehousing
and business intelligence • Database: Experience in database designing, scalabili
ty, back-up and recovery, writing and optimizing SQL code and Stored Procedures,
creating functions, views, triggers and indexes. Cloud platform: Worked on Micros
oft Azure cloud services like Document DB, SQL Azure, Stream Analytics, Event hub
, Power BI, Web Job, Web App, Power BI, Azure data lake analytics(U-SQL) Willing
to relocate anywhere WORK EXPERIENCE Software Engineer Microsoft – Bangalore, Kar
nataka January 2000 to Present 1. Microsoft Rewards Live dashboards: Description:
- Microsoft rewards is loyalty program that rewards Users for browsing and shoppi
ng online. Microsoft Rewards members can earn points when searching with Bing, br
owsing with Microsoft Edge…'
```

```
doc = nlp(text)
for ent in doc.ents:
  print(ent.text, "    ->>>>> ", ent.label_)
```

```
AI / Machine Learning     ->>>>>  Designation
Delhi     ->>>>>  Location
Microsoft     ->>>>>  Companies worked at
Software Engineer     ->>>>>  Designation
Microsoft     ->>>>>  Companies worked at
Microsoft     ->>>>>  Companies worked at
Microsoft     ->>>>>  Companies worked at
Microsoft     ->>>>>  Companies worked at
Microsoft     ->>>>>  Companies worked at
Microsoft     ->>>>>  Companies worked at
Microsoft     ->>>>>  Companies worked at
Indian Institute of Technology     ->>>>>  College Name
Machine Learning, Natural Language Processing, and Big Data Handling ADDITIONAL IN
```

Start coding or generate with AI.